

COMPARATIVE GENOMICS & THE X CHROMOSOME

WEEK 44 PCA EXERCISE



DEPARTMENT OF MOLECULAR BIOLOGY AND GENETICS

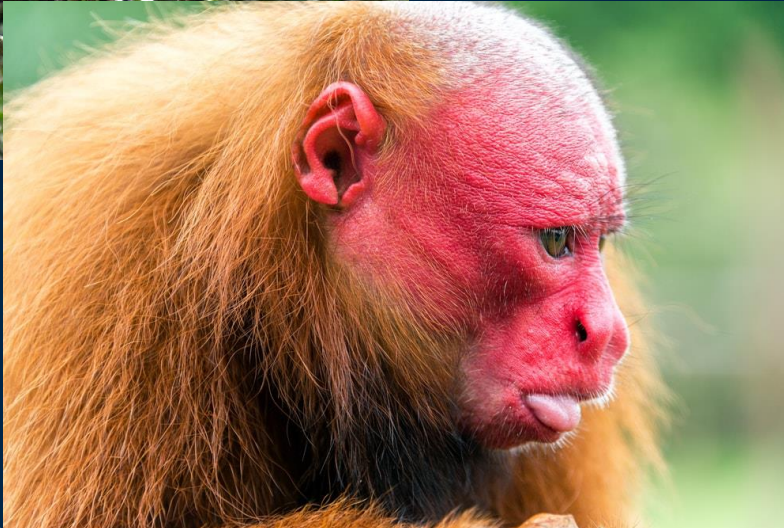
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EVOLUTIONARY THINKING
29. AUGUST 2025

EMMA FEMKE DIEPEVEEN
PH.D.-STUDERENDE



PRIMATE OF THE DAY



Bald uakari

Cacajao calvus

Occurs in South America

Bald and red head

“redness is thought to signal health”

Probably some kind of sexual selection

Females Attract mates using olfactory stimuli.



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PLAN FOR TODAY

1. Concepts important for today

- Mikkel will finish his lecture next week
- difference in mutation rate on the sex chromosomes during meiosis compared to the autosomes
- Meiotic drive

2. PCA on goats

You do not have to understand the math of a PCA now

→ focus on interpretation



DIFFERENCE IN MUTATION RATE FOR THE XY

Male meiosis occurs much more than female meiosis

→ Sperm production involves many more cell divisions than egg production throughout life → each round of division gives opportunities for replication errors or mutations

The testis is also a highly oxidative environment → results in many DNA mutations

Also the Y chromosome has nothing to recombine with, so small negative effect mutations cannot be filtered away as effectively.

Repetitive sequence content can also affect mutation spectrum

The X experiences this only in males



MEIOTIC DRIVE

when a genetic element biases its own transmission to offspring at rates $>50\%$

= **selfish** → increase their representation in the next generation at a cost to organismal fitness.

All kinds of mechanisms:

Gamete killing - Drive allele kills gametes lacking it

Fertilization advantage - Drive gametes preferentially fertilize

Post-fertilization effects - Selective embryo death

Example of the arms race:

→ X evolves allele **that kills Y sperm**

→ strong selection for Y chromosome **suppressor alleles** that resist X drive

→ Y suppressor **spread**

→ favour **new X drive** variant that escape expression

→ do the whole thing again.



MEIOTIC DRIVE AND SELECTIVE SWEEPS

- at equilibrium: multiple drive/ suppressor variants
- this was in the African populations (large, stable)
- Out Of Africa → bottleneck → Genetic drift → disrupt the balance
- earlier OOA → maintained the X drive
- main OOA → lost the X drive and also the Y suppressor
- they admixed → X drive had no opposition → spread rapidly (selective sweep)

Not proved however!!



IMPORT FOR THE EXERCISES

If you do not know how to run a jupyter notebook -> use the google link!

When installing PLINK add this:

```
%pip install pandas-plink  
  
from pandas_plink import read_plink  
import os  
  
(bim, fam, bed) = read_plink(os.getcwd() + "/data/snps")
```



NEXT WEEK

PRIMATE GENOMICS



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